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The background of the slide is a photograph of a modern urban environment. It features several multi-story buildings with glass and metal facades. One building on the left has a grid-like pattern of windows. Another building in the center has a large glass facade reflecting the sky. In the foreground, there is a city street with many bicycles parked along the curb and people walking. The sky is overcast.

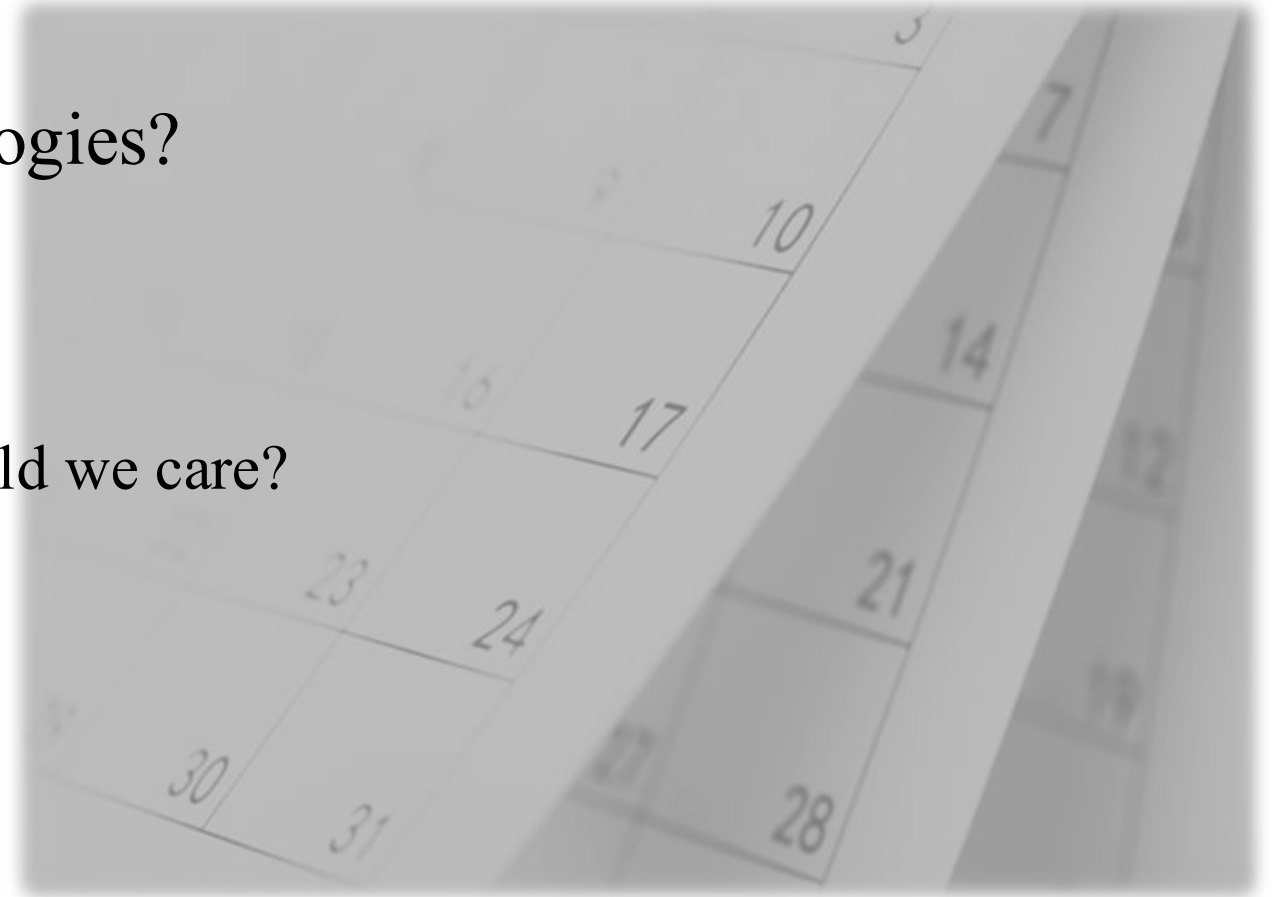
Introduction: New Technologies and Datafication

Dr. Dina Strikovic | Meeting 1



Contents

- What's new about new technologies?
 - And why should we care?
- Datafication of society
 - Who's in charge? And why should we care?
- Changes in communication
- Ethical challenges
- Exercise



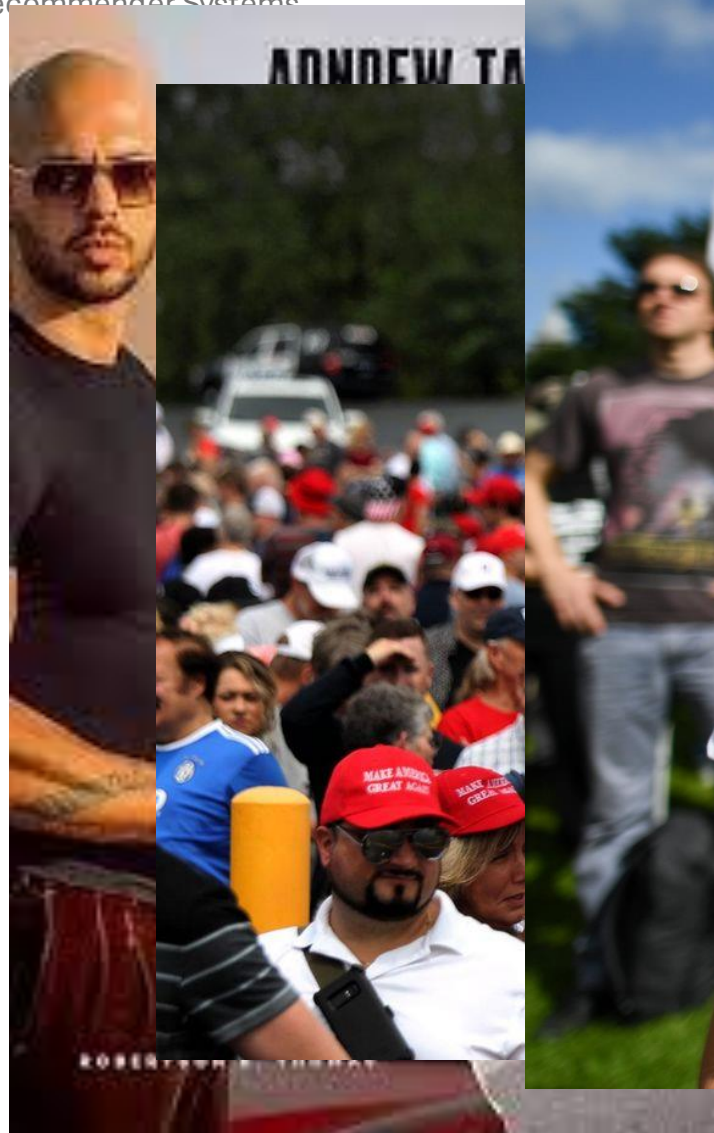


What's new about new technologies?

The benefits and challenges with digital media

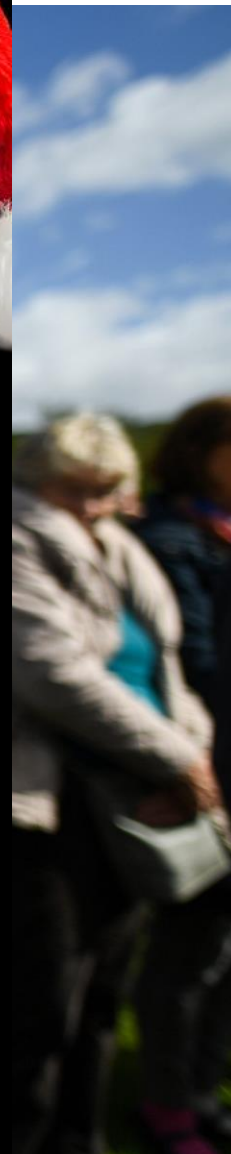
- Datafication of all (?) aspects of life!
- Decentralized publication and distribution of information
- Targeting of audiences that are most receptive
- Reciprocal → has a feeling of agency and autonomy (rightfully so?)
- Allows like-minded people to come together (or find each other)
(... unless they are already grouped together by an algorithm...)





NOT A CHRISTMAS MOVIE

DIE HARD





Information flows

Communication

Inter-personal communication has become...

- **Multi-modal:** Communicators may use different symbols, communication code systems (e.g., text, emoji, photos, videos, gifs, ...)
- **Multi-mediated:** Communication partners use a variety of platforms in their relationships (e.g., messaging apps, social media, ...)
- **Selective:** self-representation possibilities receive a boost from digital affordances



Information flows

Communication

Media has become...

- **De-centralized:** publication and distribution of information (to audiences that are most receptive)
- New **public arena:** greater pluralism detracts from inclusive agendas



How do you know what's real?

Nov 25, 2025

How personalized algorithms lead to a distorted view of reality

Study: Suggested content can lead to inaccurate generalizations



Jeff Grabmeier

Ohio State News

grabmeier.1@osu.edu





How do you know what you like?

Information about your preferences

LIFE / EVEN BETTER

The surprising thing I learned from quitting Spotify

It took months of living without Spotify's algorithm to realize I had trained it too well.

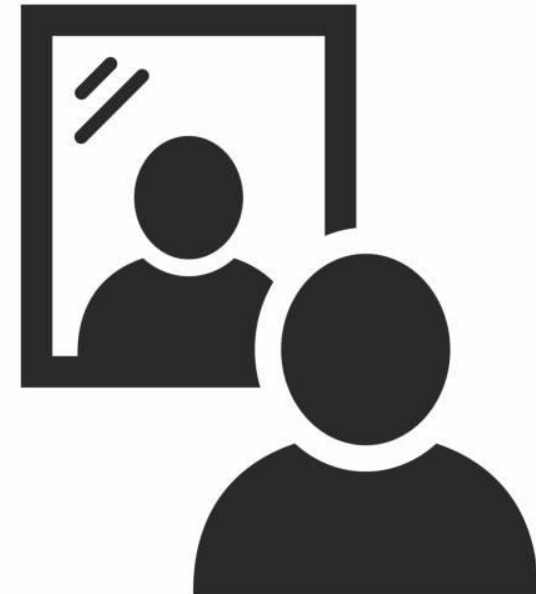
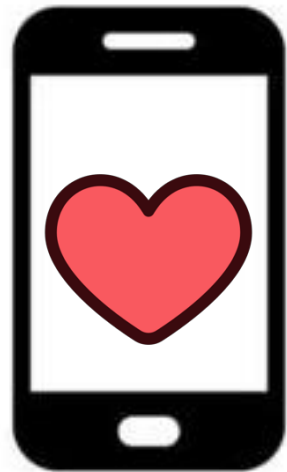
by **Adam Clark Estes**

Mar 20, 2025 at 12:30 PM GMT+1



How do you know who you are?

Information about yourself





How do you know who you are?

🕒 JULY 22, 2025

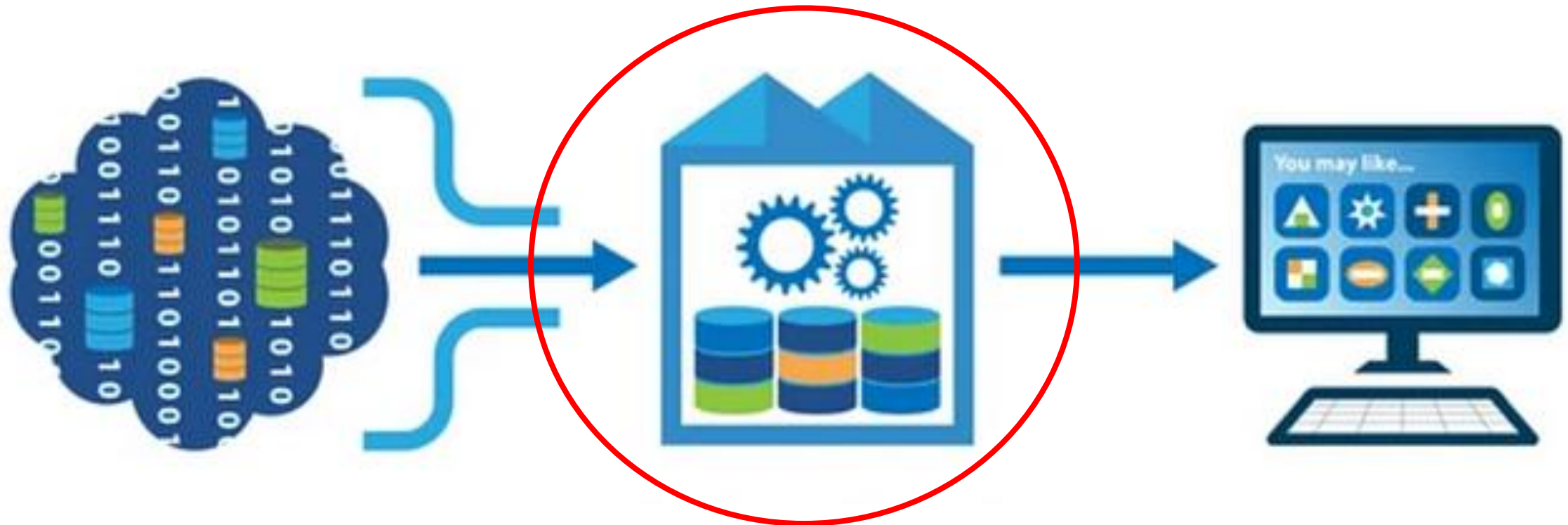
Dating app categories could be shaping you more than you know

by Kevin Gyan, The Conversation

edited by Gaby Clark, reviewed by Andrew Zinin



Editors' notes





The good

Benefits, opportunities, positive outcomes?

The bad

Challenges, problems, not ideal outcomes?

The ugly

Dangers, implications to democracy, citizens?





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Datafication



We live in a “digital society”

- **Society:** a community, nation, or broad grouping of people having common traditions, institutions, and collective activities and interests
- **Digital:** composed of data in the form of especially binary digits



We live in a “digital society”

Ubiquitous computing

- Coined by Mark Weiser
- Environment in which computational technology pervades nearly everything
- We have access to and control of this environment any time from anywhere
 - Think about the most basic settings – classroom, for example.
- At recent stages, physical world increasingly linked with electronic space



We live in a “digital society”

What makes a digital society possible?

- Pervasive digital infrastructure
 - Advanced computers (notebooks, tablets, smartphones, cars, fridges, ...)
 - Hardware as impactful as software (News, for example...)
 - Applications
 - Platforms (e.g., Facebook, Instagram, YouTube)
 - Artificial intelligence, algorithms
- Data is being generated about our behaviour → turned into recs!



Datafication

Turning something – anything – into data

- Complaining about your day 
- Reading a book 
- Listening to music 
- Attending a lecture 
- Ordering a taxi 
- Going places 

“the wider transformation of human life so that its elements can be a continual source of data”

(Mejias & Couldry, 2019, p. 2)

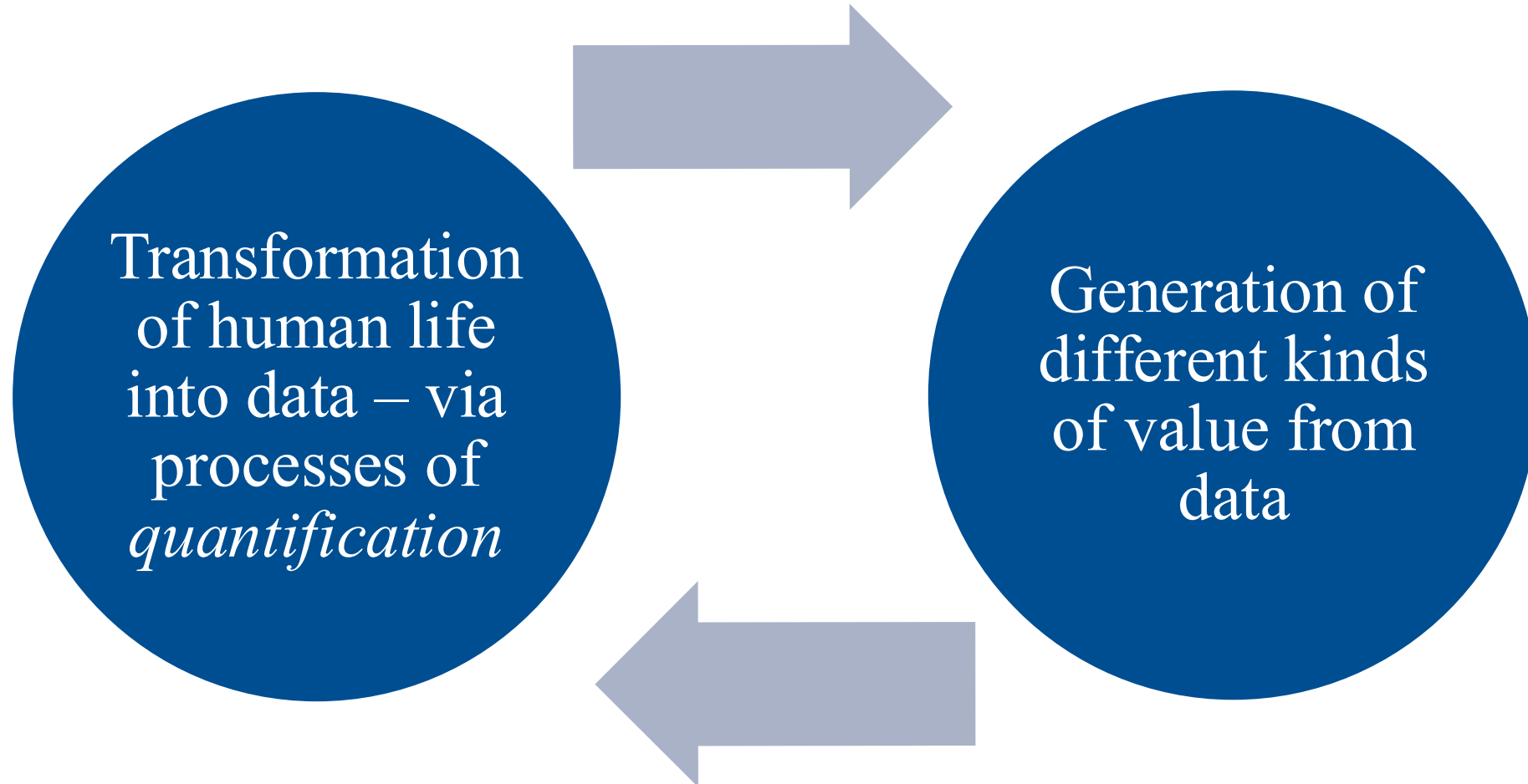


Actors and structures

- **Society at large:** Corporations, states, civic actors (e.g., NGO's, activists, journalists) or non-state actors (e.g., hacker collectives, terrorists)
- **Big corporate players**
GAFAM: *Google, Apple, Facebook, Amazon, Microsoft*
Also: *Baidu, Alibaba, Tencent, Xiaomi*
- **Social quantification sector**
Hardware, software, general platforms, data companies etc.



Process of datafication





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Power struggle



Power struggle

- **Data are not neutral:** The way in which data are produced defines what can be counted (and what counts)
- “The question of **who** is doing this codifying of life into data-fied realities acquires extreme importance at this point.” (Mejias & Couldry, 2019, p.4)

In other words:

- Who knows?
- Who decides?
- Who decides who decides?

(Zuboff, 2019)



Controversies about datafication

- As a social process: datafication linked to generation of profit
- Dependence on external, private measurements to tell us who we are, what we feel, what we should do → Human agency?
- Continuation of the *coloniality of power* (de Sousa Santos, 2016)
 - Data created by us (for free) is generating profit for companies
 - Tech we use here us based on minerals, labour coming from the global south
→ environmental, social and human cost of amplification



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Ethical Challenges



Ethical challenges

- Understanding the impact of digital technologies and digital media on the digital society is crucial – and an ethical imperative to do so
- Technologies and media do not impact only with *large changes* (e.g., new platform launched) but as a set of *day-to-day micro-decisions* (e.g., fine-tuning a news recommender algorithm)
- **You** need to be part of this discussion: As a *citizen*, as a *user*, as a *consumer*, as a *content creator*, as a *professional* (etc.)

Ethical challenges

- Not at the start (*as a binary decision*) but in every step of the process
- Developing an app
 - What does it do?
 - Who is it for?
 - What data does it use?
 - How does it learn?
 - What does it infer about the user?
 - What does it optimize for?





Ethical challenges

“engage stakeholders in order to identify, discuss and find ways of dealing with ethical issues arising from the development of new technologies, services, projects or whatever.” (Wright, 2011, p. 200)



Ethical challenges

“An ethical impact assessment must not only focus on the ethics of a technology, but on the technology itself, its *values*, how it is *perceived* and how it is *used* or *might be used* in the future, not only by itself but as *a component in a larger technological framework*.” (Wright, 2011, p. 204)



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Activity: The Algorithmic Mirror



Data Double

- **Data mapping: think about a normal day...**

When are you generating data?

- Morning?
- Commute/travel?
- Work/study?
- Social time?
- Evening/wind-down/leisure?
- Sports?

For what purpose?

- Food?
- Communication?
- Entertainment?
- Information?
- ...

What apps were you using?

What (kind of) data was generated?

Who collected it?



Data Double

Build your data double

- If a system only saw this, what would it infer from your *behaviour*?

- Interests
- Personality
- Social class
- Race
- Gender
- Occupation
-

Would this be...

- Accurate?
- Fair?
- Missing context?

What would the RS...

- show *more* of?
- show *less* of?
- suggest?



Data Double

Build your data double

- If a system only saw this, what would it infer from your *behaviour*?

- Inter
- Perso
- Social
- Race
- Gend
- Occu
-

After 6 months of this, how might this person change?

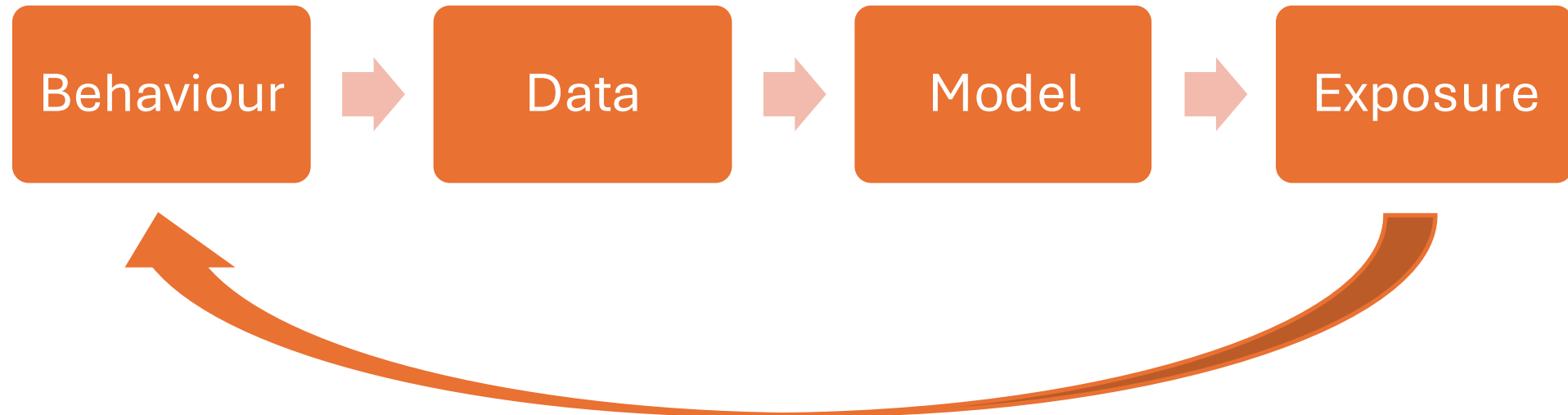
ld the RS...
more of?
ess of?
t?



Algorithmic Mirroring

Which parts of the person are...

- ... being *reflected*? ...being *produced*?





Summary

- Algorithms use data that is *not neutral* and learn from users
- Think about RS in terms of *access to* and *amplification of* information
 - ... and in terms of co-production of preferences!
 - This happens along-side micro-decisions in the development process
- Not a one-time decision, in terms of development!
 - Who are the stakeholders?
 - What are their goals?
 - Where does the data come from?
 - What is the user agency involved?
 - Where can this cycle be interrupted? Should it be interrupted?